Dr. VRINDA KARUVANCHERY

ASIC PHYSICAL DESIGN ENGINEER (EARLY CAREER) | BLOCK-LEVEL RTL-TO-GDSII | 28NM

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Early-career ASIC Physical Design Engineer with a strong VLSI academic foundation and hands-on exposure to block-level implementation. Backed by a PhD in Electronics, I bring disciplined engineering thinking, a deep understanding of physical effects, and a structured approach to reasoning about timing, parasitics, and layout-driven trade-offs. Actively seeking an individual-contributor Physical Design role in product semiconductor teams, with the goal of growing into full block ownership.

PROFESSIONAL SUMMARY

Physical Design engineer transitioning from a strong research and academic background into semiconductor product teams.Experienced in block-level physical implementation through structured industry training and independent hands-on projects. Comfortable working with report-driven analysis, iterative debugging, and engineering trade-offs, with a learning mindset aligned to early-career growth in ASIC Physical Design. Hands-on exposure to complete RTL-to-GDSII execution using both commercial (Synopsys) and open-source (OpenLANE/OpenROAD) toolchains, executed in controlled project environments to build strong flow understanding and implementation discipline.

WORK EXPERIENCE

Star VLSI

Hands-on Physical Design Projects (Self-driven, Training-aligned) (Sep 2025 - Present)

- Executed complete RTL-to-GDSII physical design flows using both commercial tool exposure and OpenLANE/OpenROAD (open-source EDA), including end-to-end implementation of the PicoRV32a RISC core within GitHub Codespaces.
- Performed report-driven iterations across synthesis, floorplanning, placement, CTS, routing, and STA, analyzing timing (WNS/TNS), clock skew, congestion, utilization, and QoR metrics to understand closure trade-offs.
- Applied ECO-style fixes including buffering, resizing, density tuning, and selective path cleanup to study timing improvement techniques and placement-driven sensitivities.
- Validated post-route timing using OpenSTA, reviewed DRC/antenna repair status, and inspected PDN (VDD/VSS) topology to ensure physical integrity and signoff-level correctness.
- Documented flow execution, report interpretation, and debugging methodology, reinforcing structured implementation reasoning aligned with industry PD practices.

PES University

ASSISTANT PROFESSOR (Bangalore, India) (Sep 2024 - Feb 2025)

- Taught core Electronics and VLSI subjects, maintaining strong hands-on engagement with circuit behavior, devices, and system-level design concepts.
- Guided students on industry-aligned technical projects, reinforcing applied engineering thinking alongside fundamentals.

Basingstoke College of Engineering

SESSIONAL LECTURER (London, United Kingdom) (Aug 2023 - Jun 2024)

- Delivered structured technical instruction aligned with applied engineering outcomes.
- Mentored students on project-based learning emphasizing correctness, rigor, and execution.

ChipEdge Technologies Pvt Ltd

PHYSICAL DESIGN TRAINEE (Bangalore, India) (Jan 2023 - Jul 2023)

- Trained on block-level RTL-to-GDSII implementation for semi-custom ASICs using Synopsys Design Compiler, ICC2, PrimeTime, and StarRC, following industry-standard sequencing.
- Executed floorplanning, placement, routing, and clock tree synthesis with focus on timing-driven implementation, congestion awareness, and structured closure methodology.
- Performed static timing analysis (PrimeTime) including setup/hold checks and MCMM fundamentals, interpreting reports to identify critical paths and closure bottlenecks.
- Applied controlled ECO iterations (buffering, cell resizing, incremental CTS refinement) to study timing convergence behavior under 28nm libraries.
- Gained exposure to 14nm FinFET concepts, developing awareness of parasitics, clock skew, and power–timing trade-offs.
- Used basic TCL scripting to automate PD flow execution and extract structured timing and QoR reports.

Amrita Vishwa Vidya Peetham

RESEARCH SCHOLAR (Bangalore, India) (Oct 2015 - Dec 2022)

- Conducted advanced research in RF integrated circuits and passive device modeling, strengthening understanding of layout-dependent behavior and parasitic effects.
- Executed 500+ hours of electromagnetic simulations using Ansys HFSS, reinforcing physical intuition relevant to signal integrity and high-frequency effects.
- Published 4 peer-reviewed journal papers and presented at 5 international conferences in semiconductor-focused forums.
- Developed strong analytical, documentation, and problem-decomposition skills directly applicable to design debug and root-cause analysis.

T.John College of Engineering

ASSISTANT PROFESSOR (Bangalore, India) (Aug 2013 - Jul 2014)

- Delivered engaging lectures on Microwave Devices and Electrical Circuits, integrating interactive teaching techniques.
- Played a key role in curriculum refinement to enhance student comprehension and academic performance.

- Led STEM

SKILLS

Technical Skills:
ASIC Physical Design: Floorplanning, Placement & Routing (PnR), Clock Tree Synthesis (CTS), Static Timing Analysis (STA), Timing Closure Fundamentals, Multi-Corner Multi-Mode (MCMM) Analysis,
Timing & Physical Effects: Setup/Hold Analysis, Signal Integrity Awareness, Power Optimization Basics,
Technology & Design Context: 28nm CMOS Process Technology, Hierarchical and Block-Level Design Flows, Low-Power Design Awareness (Clock Gating, Power Intent Concepts),
Scripting & Automation: TCL (basic scripting for PD flow execution and report analysis), Shell Scripting (introductory automation exposure),
Simulation & Analysis: Ansys HFSS (electromagnetic and parasitic modeling),
RTL-to-GDSII Full-Flow Execution: OpenLANE/OpenROAD-based implementation, synthesis-to-signoff validation, congestion and utilization analysis, PDN inspection, DRC/antenna awareness,,
EDA Tools: Synopsys IC Compiler II (ICC2), PrimeTime, StarRC, Design Compiler; Cadence Innovus, Cadence Virtuoso; Mentor Graphics Calibre (DRC, LVS), OpenLANE, OpenROAD, OpenSTA, Magic, KLayout

Soft Skills: Problem-Solving and Analytical Thinking, Clear Technical Communication and Documentation,
Collaboration with Cross-Functional Engineering Teams, Strong Learning Agility and Coachability,
Attention to Detail and Engineering Discipline, Time Management and Task Prioritization,
Professional Mentorship and Knowledge Sharing, Adaptability in Fast-Paced Product Development Environments

Core Competencies: Block-Level Physical Design Implementation (Floorplan to STA),
Understanding of Timing, Parasitics, and Layout-Driven Trade-offs,
ASIC Design Flow Familiarity from RTL Handoff to Physical Sign-off (Conceptual to Block-Level),
Design Debugging and Root-Cause Analysis Using Structured Engineering Approaches,
Applying Strong VLSI Fundamentals to Practical PD Execution,
Documentation and Design Review Readiness in Engineering Environments,
Ability to Learn, Adapt, and Grow into Block Ownership within Product Teams,
Ability to execute and document end-to-end RTL-to-GDSII flows with structured report interpretation and timing-driven debugging discipline

EDUCATION

University of Calicut, India (2006 - 2010)
Bachelors Electronics and Communication Engineering

Amrita University, India (2011 - 2013)
Masters VLSI Design Engineering

Amrita University, India (Oct 2015 - Dec 2022)
PhD Passive Device Modeling for RFICs

ACHIEVEMENTS

- Published 4 high-impact research papers on electromagnetic modeling in esteemed international journals.
- Presented at 5 international conferences, showcasing cutting-edge research in RF integrated circuits.
- Designed and revitalized VLSI and microwave devices curricula, leading to a measurable improvement in student performance and understanding.
- Increased collaborative research output by 25% through effective mentorship and leadership.

HOBBIES

Circuit Design, Technical Blogging, Educational Technology

LANGUAGES

English, Hindi, Malayalam